

UPTET

Top 50 Most Repeated Previous Year Questions

Part 1: Mathematics (25 Questions)

1. A sum of money doubles itself in 10 years at simple interest. What is the rate of interest?
A) 5% B) 10% C) 15% D) 20%
2. The HCF and LCM of two numbers are 12 and 144 respectively. If one number is 36, find the other. A) 24 B) 48 C) 72 D) 120
3. What is the value of $(x - y)^3 + (y - z)^3 + (z - x)^3$? A) 0 B) $3(x-y)(y-z)(z-x)$ C) 1 D) $x^3 + y^3 + z^3$
4. In a right-angled triangle, if the hypotenuse is 13 cm and one side is 5 cm, the third side is? A) 8 cm B) 10 cm C) 12 cm D) 14 cm
5. Find the unit digit in the product $(2467)^{153}$. A) 1 B) 3 C) 7 D) 9
6. If $a/b = 2/3$ and $b/c = 4/5$, then find the ratio of a:b:c. A) 8:12:15 B) 2:3:5 C) 8:15:12 D) 4:6:5
7. The sum of the interior angles of a regular hexagon is: A) 540° B) 720° C) 900° D) 1080°
8. The perimeter of a rectangle is 40 cm. If its length is 12 cm, find its area. A) 96 cm^2 B) 80 cm^2 C) 144 cm^2 D) 120 cm^2
9. A person buys a cycle for Rs 1400 and sells it at a loss of 15%. What is the selling price?
A) Rs 1190 B) Rs 1200 C) Rs 1150 D) Rs 1100
10. The mean of the first 10 prime numbers is: A) 11.5 B) 12.9 C) 13.5 D) 14.1
11. Two supplementary angles are in the ratio 2:3. Find the largest angle. A) 72° B) 90° C) 108° D) 120°
12. If the radius of a circle is increased by 50%, its area will increase by: A) 50% B) 100% C) 125% D) 225%
13. The square root of 0.000441 is: A) 0.21 B) 0.021 C) 0.0021 D) 0.00021
14. If $x + 1/x = 5$, find the value of $x^2 + 1/x^2$. A) 23 B) 25 C) 27 D) 29
15. The volume of a cube is 216 cm^3 . Find its total surface area. A) 144 cm^2 B) 196 cm^2 C) 216 cm^2 D) 256 cm^2
16. Which of the following is the smallest prime number? A) 0 B) 1 C) 2 D) 3
17. What percent of 1 day is 36 minutes? A) 2.5% B) 3.5% C) 4.5% D) 5.5%

18. The angles of a triangle are in the ratio 1:2:3. Find the largest angle. A) 30° B) 60° C) 90°
D) 120°
19. A train 150m long is running at 72 km/h. How long will it take to cross a pole? A) 5.5 sec
B) 7.5 sec C) 8.5 sec D) 10.5 sec
20. In a class of 50 students, 30 are boys. What is the percentage of girls? A) 20% B) 40% C)
50% D) 60%
21. Find the value of $0.333... + 0.666...$ A) 0.999 B) 1.0 C) 0 D) None of the above
22. Find the median of the data set: 2, 4, 6, 8, 10. A) 4 B) 6 C) 8 D) 10
23. If $3^x = 81$, find the value of x. A) 2 B) 3 C) 4 D) 5
24. Evaluate: 55% of 1000 + 45% of 2000. A) 1250 B) 1350 C) 1450 D) 1550
25. What is the total number of factors of 24? A) 4 B) 6 C) 8 D) 10

Part 2: Science (25 Questions)

26. **What is the SI unit of Force?** A) Joule B) Newton C) Watt D) Pascal
27. **Scurvy is caused by the deficiency of which vitamin?** A) Vitamin A B) Vitamin B C) Vitamin C D) Vitamin D
28. **Which gas is most abundant in the Earth's atmosphere?** A) Oxygen B) Carbon Dioxide C) Nitrogen D) Hydrogen
29. **Which organelle is known as the powerhouse of the cell?** A) Nucleus B) Ribosome C) Mitochondria D) Golgi apparatus
30. **The chemical formula for baking soda is:** A) Na_2CO_3 B) NaHCO_3 C) NaCl D) NaOH
31. **Which of the following metals is liquid at room temperature?** A) Sodium B) Mercury C) Iron D) Aluminum
32. **A 'Light Year' is a unit of measurement for:** A) Time B) Speed C) Distance D) Intensity of light
33. **The process by which plants synthesize their own food is called:** A) Respiration B) Transpiration C) Photosynthesis D) Osmosis
34. **Which blood group is universally known as the universal donor?** A) A B) B C) AB D) O
35. **The normal body temperature of a healthy human is approximately:** A) 35°C B) 37°C C) 39°C D) 41°C
36. **Sound travels fastest in which of the following mediums?** A) Vacuum B) Air C) Water D) Solid (like Steel)
37. **The lens naturally present in the human eye is:** A) Concave B) Convex C) Cylindrical D) Bifocal
38. **Which acid is predominantly found in lemons and citrus fruits?** A) Acetic acid B) Citric acid C) Lactic acid D) Tartaric acid
39. **The hardest naturally occurring substance on Earth is:** A) Gold B) Iron C) Diamond D) Graphite
40. **Which planet in our solar system is known as the Red Planet?** A) Venus B) Mars C) Jupiter D) Saturn
41. **The gas essential for human respiration is:** A) Carbon dioxide B) Nitrogen C) Oxygen D) Helium
42. **An ammeter is an instrument used to measure:** A) Voltage B) Electric Current C) Resistance D) Power
43. **Deficiency of iron in the human diet primarily causes:** A) Goiter B) Anemia C) Rickets D) Night blindness

44. Which part of the plant is responsible for the transport of water from the roots to the leaves? A) Phloem B) Xylem C) Stomata D) Epidermis
45. A convex mirror is generally used as a: A) Dentist's mirror B) Makeup mirror C) Rear-view mirror in vehicles D) Shaving mirror
46. The pH value of pure water at room temperature is: A) 0 B) 7 C) 14 D) 10
47. What is the SI unit of Electric Current? A) Volt B) Ampere C) Ohm D) Coulomb
48. Which vitamin is synthesized in the human body in the presence of sunlight? A) Vitamin A B) Vitamin K C) Vitamin E D) Vitamin D
49. What is the main component of Biogas? A) Ethane B) Methane C) Propane D) Butane
50. The filament of an electric bulb is usually made of: A) Copper B) Aluminum C) Tungsten D) Lead